

Sliding-window

Array/String K

Apply

→ output → subarray/substring

1-12

① Fixed window

② Variable size window

Fixed size

[2, 3, 5, 1, 7, 9, 4, 2, 3, 1] K=3

$\frac{n \times k}{T.C}$

[2, 3, 5, 1, 7, 9, 4, 2, 3, 1]

10+1=11-2
=9+7-3
=13+9-5
=17+4-1
=20+2-7
=15+3-7
=11-10

```
public static int Maximum_SubArray_Sum(int [] arr,int k) {  
    // 1st window  
    int ans=0,sum=0;  
    for (int i = 0; i <k; i++) {  
        sum+=arr[i];  
    }  
    ans=sum;  
    for (int i = k; i < arr.length; i++) {  
        sum+=arr[i]; // window grow  
        sum-=arr[i-k]; //window shrink  
        ans=Math.max(ans, sum); // ans calculate  
    }  
    return ans;  
}
```

1 → [2] ✓
2 { [1]
[2,1]
3 { [3]
[1,3]
[2,1,3]
1 [4]
2 { [2]
[4,2]
3 { [4]
[2,4]
[1,4]
4 [1]
5 [2]
6 [3]
7 [4]
8 [1]
9 [2]
10 [3]
11 [4]
12 [1]

[2, 3, 5, 1, 7, 9, 4, 2, 3, 1]

si=12
ei=19
19-12+1

ans = 1+2+3+4+5+6+7+8+9+10+11+12+13+14+15+16+17+18+19+20+21+22+23+24+25+26+27+28+29+30+31+32+33+34+35+36+37+38+39+40+41+42+43+44+45+46+47+48+49+50+51+52+53+54+55+56+57+58+59+60+61+62+63+64+65+66+67+68+69+70+71+72+73+74+75+76+77+78+79+80+81+82+83+84+85+86+87+88+89+90+91+92+93+94+95+96+97+98+99+100+101+102+103+104+105+106+107+108+109+110+111+112+113+114+115+116+117+118+119+120+121+122+123+124+125+126+127+128+129+130+131+132+133+134+135+136+137+138+139+140+141+142+143+144+145+146+147+148+149+150+151+152+153+154+155+156+157+158+159+160+161+162+163+164+165+166+167+168+169+170+171+172+173+174+175+176+177+178+179+180+181+182+183+184+185+186+187+188+189+190+191+192+193+194+195+196+197+198+199+200+201+202+203+204+205+206+207+208+209+210+211+212+213+214+215+216+217+218+219+220+221+222+223+224+225+226+227+228+229+230+231+232+233+234+235+236+237+238+239+240+241+242+243+244+245+246+247+248+249+250+251+252+253+254+255+256+257+258+259+260+261+262+263+264+265+266+267+268+269+270+271+272+273+274+275+276+277+278+279+280+281+282+283+284+285+286+287+288+289+290+291+292+293+294+295+296+297+298+299+300+301+302+303+304+305+306+307+308+309+310+311+312+313+314+315+316+317+318+319+320+321+322+323+324+325+326+327+328+329+330+331+332+333+334+335+336+337+338+339+340+341+342+343+344+345+346+347+348+349+350+351+352+353+354+355+356+357+358+359+360+361+362+363+364+365+366+367+368+369+370+371+372+373+374+375+376+377+378+379+380+381+382+383+384+385+386+387+388+389+390+391+392+393+394+395+396+397+398+399+400+401+402+403+404+405+406+407+408+409+410+411+412+413+414+415+416+417+418+419+420+421+422+423+424+425+426+427+428+429+430+431+432+433+434+435+436+437+438+439+440+441+442+443+444+445+446+447+448+449+450+451+452+453+454+455+456+457+458+459+460+461+462+463+464+465+466+467+468+469+470+471+472+473+474+475+476+477+478+479+480+481+482+483+484+485+486+487+488+489+490+491+492+493+494+495+496+497+498+499+500+501+502+503+504+505+506+507+508+509+510+511+512+513+514+515+516+517+518+519+520+521+522+523+524+525+526+527+528+529+530+531+532+533+534+535+536+537+538+539+540+541+542+543+544+545+546+547+548+549+550+551+552+553+554+555+556+557+558+559+560+561+562+563+564+565+566+567+568+569+570+571+572+573+574+575+576+577+578+579+580+581+582+583+584+585+586+587+588+589+590+591+592+593+594+595+596+597+598+599+600+601+602+603+604+605+606+607+608+609+610+611+612+613+614+615+616+617+618+619+620+621+622+623+624+625+626+627+628+629+630+631+632+633+634+635+636+637+638+639+640+641+642+643+644+645+646+647+648+649+650+651+652+653+654+655+656+657+658+659+660+661+662+663+664+665+666+667+668+669+670+671+672+673+674+675+676+677+678+679+680+681+682+683+684+685+686+687+688+689+690+691+692+693+694+695+696+697+698+699+700+701+702+703+704+705+706+707+708+709+710+711+712+713+714+715+716+717+718+719+720+721+722+723+724+725+726+727+728+729+730+731+732+733+734+735+736+737+738+739+740+741+742+743+744+745+746+747+748+749+750+751+752+753+754+755+756+757+758+759+760+761+762+763+764+765+766+767+768+769+770+771+772+773+774+775+776+777+778+779+780+781+782+783+784+785+786+787+788+789+790+791+792+793+794+795+796+797+798+799+800+801+802+803+804+805+806+807+808+809+810+811+812+813+814+815+816+817+818+819+820+821+822+823+824+825+826+827+828+829+830+831+832+833+834+835+836+837+838+839+840+841+842+843+844+845+846+847+848+849+850+851+852+853+854+855+856+857+858+859+860+861+862+863+864+865+866+867+868+869+870+871+872+873+874+875+876+877+878+879+880+881+882+883+884+885+886+887+888+889+890+891+892+893+894+895+896+897+898+899+900+901+902+903+904+905+906+907+908+909+910+911+912+913+914+915+916+917+918+919+920+921+922+923+924+925+926+927+928+929+930+931+932+933+934+935+936+937+938+939+940+941+942+943+944+945+946+947+948+949+950+951+952+953+954+955+956+957+958+959+960+961+962+963+964+965+966+967+968+969+970+971+972+973+974+975+976+977+978+979+980+981+982+983+984+985+986+987+988+989+990+991+992+993+994+995+996+997+998+999+1000+1001+1002+1003+1004+1005+1006+1007+1008+1009+1010+1011+1012+1013+1014+1015+1016+1017+1018+1019+1020+1021+1022+1023+1024+1025+1026+1027+1028+1029+1030+1031+1032+1033+1034+1035+1036+1037+1038+1039+1040+1041+1042+1043+1044+1045+1046+1047+1048+1049+1050+1051+1052+1053+1054+1055+1056+1057+1058+1059+1060+1061+1062+1063+1064+1065+1066+1067+1068+1069+1070+1071+1072+1073+1074+1075+1076+1077+1078+1079+1080+1081+1082+1083+1084+1085+1086+1087+1088+1089+1090+1091+1092+1093+1094+1095+1096+1097+1098+1099+1100+1101+1102+1103+1104+1105+1106+1107+1108+1109+1110+1111+1112+1113+1114+1115+1116+1117+1118+1119+1120+1121+1122+1123+1124+1125+1126+1127+1128+1129+1130+1131+1132+1133+1134+1135+1136+1137+1138+1139+1140+1141+1142+1143+1144+1145+1146+1147+1148+1149+1150+1151+1152+1153+1154+1155+1156+1157+1158+1159+1160+1161+1162+1163+1164+1165+1166+1167+1168+1169+1170+1171+1172+1173+1174+1175+1176+1177+1178+1179+1180+1181+1182+1183+1184+1185+1186+1187+1188+1189+1190+1191+1192+1193+1194+1195+1196+1197+1198+1199+1200+1201+1202+1203+1204+1205+1206+1207+1208+1209+1210+1211+1212+1213+1214+1215+1216+1217+1218+1219+1220+1221+1222+1223+1224+1225+1226+1227+1228+1229+1230+1231+1232+1233+1234+1235+1236+1237+1238+1239+1240+1241+1242+1243+1244+1245+1246+1247+1248+1249+1250+1251+1252+1253+1254+1255+1256+1257+1258+1259+1260+1261+1262+1263+1264+1265+1266+1267+1268+1269+1270+1271+1272+1273+1274+1275+1276+1277+1278+1279+1280+1281+1282+1283+1284+1285+1286+1287+1288+1289+1290+1291+1292+1293+1294+1295+1296+1297+1298+1299+1300+1301+1302+1303+1304+1305+1306+1307+1308+1309+1310+1311+1312+1313+1314+1315+1316+1317+1318+1319+1320+1321+1322+1323+1324+1325+1326+1327+1328+1329+1330+1331+1332+1333+1334+1335+1336+1337+1338+1339+1340+1341+1342+1343+1344+1345+1346+1347+1348+1349+1350+1351+1352+1353+1354+1355+1356+1357+1358+1359+1360+1361+1362+1363+1364+1365+1366+1367+1368+1369+1370+1371+1372+1373+1374+1375+1376+1377+1378+1379+1380+1381+1382+1383+1384+1385+1386+1387+1388+1389+1390+1391+1392+1393+1394+1395+1396+1397+1398+1399+1400+1401+1402+1403+1404+1405+1406+1407+1408+1409+1410+1411+1412+1413+1414+1415+1416+1417+1418+1419+1420+1421+1422+1423+1424+1425+1426+1427+1428+1429+1430+1431+1432+1433+1434+1435+1436+1437+1438+1439+1440+1441+1442+1443+1444+1445+1446+1447+1448+1449+1450+1451+1452+1453+1454+1455+1456+1457+1458+1459+1460+1461+1462+1463+1464+1465+1466+1467+1468+1469+1470+1471+1472+1473+1474+1475+1476+1477+1478+1479+1480+1481+1482+1483+1484+1485+1486+1487+1488+1489+1490+1491+1492+1493+1494+1495+1496+1497+1498+1499+1500+1501+1502+1503+1504+1505+1506+1507+1508+1509+1510+1511+1512+1513+1514+1515+1516+1517+1518+1519+1520+1521+1522+1523+1524+1525+1526+1527+1528+1529+1530+1531+1532+1533+1534+1535+1536+1537+1538+1539+1540+1541+1542+1543+1544+1545+1546+1547+1548+1549+1550+1551+1552+1553+1554+1555+1556+1557+1558+1559+1560+1561+1562+1563+1564+1565+1566+1567+1568+1569+1570+1571+1572+1573+1574+1575+1576+1577+1578+1579+1580+1581+1582+1583+1584+1585+1586+1587+1588+1589+1590+1591+1592+1593+1594+1595+1596+1597+1598+1599+1600+1601+1602+1603+1604+1605+1606+1607+1608+1609+1610+1611+1612+1613+1614+1615+1616+1617+1618+1619+1620+1621+1622+1623+1624+1625+1626+1627+1628+1629+1630+1631+1632+1633+1634+1635+1636+1637+1638+1639+1640+1641+1642+1643+1644+1645+1646+1647+1648+1649+1650+1651+1652+1653+1654+1655+1656+1657+1658+1659+1660+1661+1662+1663+1664+1665+1666+1667+1668+1669+1670+1671+1672+1673+1674+1675+1676+1677+1678+1679+1680+1681+1682+1683+1684+1685+1686+1687+1688+1689+1690+1691+1692+1693+1694+1695+1696+1697+1698+1699+1700+1701+1702+1703+1704+1705+1706+1707+1708+1709+1710+1711+1712+1713+1714+1715+1716+1717+1718+1719+1720+1721+1722+1723+1724+1725+1726+1727+1728+1729+1730+1731+1732+1733+1734+1735+1736+1737+1738+1739+1740+1741+1742+1743+1744+1745+1746+1747+1748+1749+1750+1751+1752+1753+1754+1755+1756+1757+1758+1759+1760+1761+1762+1763+1764+1765+1766+1767+1768+1769+1770+1771+1772+1773+1774+1775+1776+1777+1778+1779+1780+1781+1782+1783+1784+1785+1786+1787+1788+1789+1790+1791+1792+1793+1794+1795+1796+1797+1798+1799+1800+1801+1802+1803+1804+1805+1806+1807+1808+1809+1810+1811+1812+1813+1814+1815+1816+1817+1818+1819+1820+1821+1822+1823+1824+1825+1826+1827+1828+1829+1830+1831+1832+1833+1834+1835+1836+1837+1838+1839+1840+1841+1842+1843+1844+1845+1846+1847+1848+1849+1850+1851+1852+1853+1854+1855+1856+1857+1858+1859+1860+1861+1862+1863+1864+1865+1866+1867+1868+1869+1870+1871+1872+1873+1874+1875+1876+1877+1878+1879+1880+1881+1882+1883+1884+1885+1886+1887+1888+1889+1890+1891+1892+1893+1894+1895+1896+1897+1898+1899+1900+1901+1902+1903+1904+1905+1906+1907+1908+1909+1910+1911+1912+1913+1914+1915+1916+1917+1918+1919+1920+1921+1922+1923+1924+1925+1926+1927+1928+1929+1930+1931+1932+1933+1934+1935+1936+1937+1938+1939+1940+1941+1942+1943+1944+1945+1946+1947+1948+1949+1950+1951+1952+1953+1954+1955+1956+1957+1958+1959+1960+1961+1962+1963+1964+1965+1966+1967+1968+1969+1970+1971+1972+1973+1974+1975+1976+1977+1978+1979+1980+1981+1982+1983+1984+1985+1986+1987+1988+1989+1990+1991+1992+1993+1994+1995+1996+1997+1998+1999+2000+2001+2002+2003+2004+2005+2006+2007+2008+2009+2010+2011+2012+2013+2014+2015+2016+2017+2018+2019+2020+2021+2022+2023+2024+2025+2026+2027+2028+2029+2030+2031+2032+2033+2034+2035+2036+2037+2038+2039+2040+2041+2042+2043+2044+2045+2046+2047+2048+2049+2050+2051+2052+2053+2054+2055+2056+2057+2058+2059+2060+2061+2062+2063+2064+2065+2066+2067+2068+2069+2070+2071+2072+2073+2074+2075+2076+2077+2078+2079+2080+2081+2082+2083+2084+2085+2086+2087+2088+2089+2090+2091+2092+2093+2094+2095+2096+2097+2098+2099+2100+2101+2102+2103+2104+2105+2106+2107+2108+2109+2110+2111+2112+2113+2114+2115+2116+2117+2118+2119+2120+2121+2122+2123+2124+2125+2126+2127+2128+2129+2130+2131+2132+2133+2134+2135+2136+2137+2138+2139+2140+2141+2142+2143+2144+2145+2146+2147+2148+2149+2150+2151+2152+2153+2154+2155+2156+2157+2158+2159+2160+2161+2162+2163+2164+2165+2166+2167+2168+2169+2170+2171+2172+2173+2174+2175+2176+2177+2178+2179+2180+2181+2182+2183+2184+2185+2186+2187+2188+2189+2190+2191+2192+2193+2194+2195+2196+2197+2198+2199+2200+2201+2202+2203+2204+2205+2206+2207+2208+2209+2210+2211+2212+2213+2214+2215+2216+2217+2218+2219+2220+2221+2222+2223+2224+2225+2226+2227+2228+2229+2230+2231+2232+2233+2234+2235+2236+2237+2238+2239+2240+2241+2242+2243+2244+2245+2246+2247+2248+2249+2250+2251+2252+2253+2254+2255+2256+2257+2258+2259+2260+2261+2262+2263+2264+2265+2266+2267+2268+2269+2270+2271+2272+2273+2274+2275+2276+2277+2278+2279+2280+2281+2282+2283+2284+2285+2286+2287+2288+2289+2290+2291+2292+2293+2294+2295+2296+2297+2298+2299+2300+2301+2302+2303+2304+2305+2306+2307+2308+2309+2310+2311+2312+2313+2314+2315+2316+2317+2318+2319+2320+2321+2322+2323+2324+2325+2326+2327+2328+2329+2330+2331+2332+2333+2334+2335+2336+2337+2338+2339+2340+2341+2342+2343+2344+2345+2346+2347+2348+2349+2350+2351+2352+2353+2354+2355+2356+2357+2358+2359+2360+2361+2362+2363+2364+2365+2366+2367+2368+2369+2370+2371+2372+2373+2374+2375+2376+2377+2378+2379+2380+2381+2382+2383+2384+2385+2386+2387+2388+2389+2390+2391+2392+2393+2394+2395+2396+2397+2398+2399+2400+2401+2402+2403+2404+2405+2406+2407+2408+2409+2410+2411+2412+2413+2414+2415+2416+2417+2418+2419+2420+2421+2422+2423+2424+2425+2426+2427+2428+2429+2430+2431+2432+2433+2434+2435+2436+2437+2438+2439+2440+2441+2442+2443+2444+2445+2446+2447+2448+2449+2450+2451+2452+2453+2454+2455+2456+2457+2458+2459+2460+2461+2462+2463+2464+2465+2466+2467+2468+2469+2470+2471+2472+2473+2474+2475+2476+2477+2478+2479+2480+248